

Antonio Lara Gutiérrez

PhD Candidate & FPU Predoctoral Researcher. AI for Cybersecurity
NICS Lab. Dept. of Languages and Computer Science. ETSI Informática, University of Málaga

📍 Málaga, Spain | 📞 +34 628 78 73 78 | ✉ antoniolaragutierrez2000@gmail.com
🐙 github.com/antoniol00 | 🔗 linkedin.com/in/antonio-lara-gutierrez

RESEARCH PROFILE

PhD candidate and FPU predoctoral researcher at the University of Málaga (NICS Lab). My research lies at the intersection of **Artificial Intelligence and Cybersecurity**, with emphasis on: (i) **concept-drift detection and adaptation** in AI-driven threat detection systems; (ii) **multimodal malware classification and attribution** through static, dynamic and visual feature fusion; (iii) **generative models**—GANs and LLMs—for adversarial robustness of intrusion detection and for adaptive cyber deception; and (iv) **LLM agents** for automated threat intelligence. My goal is to build robust, verifiable and maintainable AI defenses against advanced cyber threats.

EDUCATION

Sep. 2024 – Present	PhD in Information Technologies <i>University of Málaga — NICS Lab</i>	Málaga, Spain
	▶ Research focus: AI-based malware analysis and attribution, GAN-driven proactive defense, concept-drift adaptation and threat intelligence. ▶ Supervisors: Dr. Carmen Fernández-Gago and Dr. Jose A. Onieva. ▶ Funded by an FPU predoctoral contract (Spanish Ministry of Science, Innovation and Universities).	
Jul. 2022 – Jun. 2024	M.Sc. in Computer Engineering — Cybersecurity (90 ECTS) <i>University of Málaga</i>	Málaga, Spain
	▶ GPA 9.56 / 10. 5 courses with Distinction (22.5 ECTS). Ranked 1st in cohort (Extraordinary Graduation Award). ▶ Thesis: <i>Dynamic Risk Management and Resilience in Industrial Environments using Artificial Intelligence</i> — 10/10, Outstanding Thesis Award.	
Jul. 2018 – Jul. 2022	B.Sc. in Computer Engineering — Information Technologies (240 ECTS) <i>University of Málaga</i>	Málaga, Spain
	▶ GPA 9.29 / 10. 22 courses with Distinction (138 ECTS). Ranked 1st in cohort (Extraordinary Graduation Award). ▶ Thesis: <i>Traceability of Control Actions in EV Charging Stations via Mobile App and Blockchain</i> — 10/10 (Distinction), Outstanding Thesis Award.	
Oct. 2023 – Oct. 2024	Master's in Big Data and Business Analytics <i>School of Industrial Organisation (EOI)</i>	Madrid, Spain
	Data-driven strategy, Business Intelligence, Big Data technologies and advanced analytics.	

PUBLICATIONS

Own name in bold. Impact factor (IF) and quartile from Journal Citation Reports (JCR).

Journal Articles (JCR-indexed)

[J1] **Lara-Gutiérrez, A.**, Fernández-Gago, C. & Onieva, J. A. “**A framework for drift detection and adaptation in AI-driven anomaly and threat detection systems.**” *International Journal of Information Security* (Springer), 24, 199, 2025.
 **Q1**  **IF 5.0**  doi:10.1007/s10207-025-01118-9

[J2] **Lara-Gutiérrez, A.**, Fernández-Gago, C. & Onieva, J. A. “**Towards maintainable AI-driven network anomaly and threat detection: a comparative analysis of datasets, preprocessing techniques, and model trade-offs.**” *Artificial Intelligence Review* (Springer), 2026.
 **Q1**  **IF 18.8**  doi:10.1007/s10462-026-11667-7

- [J3] Lara-Gutiérrez, A., Fernández-Gago, C. & Onieva, J. A. “Adversarial red teaming and imbalance correction in heterogeneous NIDS: a class-specific adaptive GAN framework.” *Journal of Information Security and Applications* (Elsevier), 102, 104584, 2026.
JCR Q2 IF 4.4  doi:10.1016/j.jisa.2026.104584
- [J4] Lara-Gutiérrez, A., Zamorano, J. & Onieva, J. A. “GenPot: a generative honeypot architecture for adaptive web and API interaction.” *Applied Intelligence* (Springer), 56, 419, 2026.
JCR Q2 IF 3.5  doi:10.1007/s10489-026-07442-4

Under Review

- [R1] Lara-Gutiérrez, A., Álvarez, N. & Onieva, J. A. “Multimodal malware family classification: a large-scale study on early vs. late fusion of static, dynamic, and visual features.” *Journal of Network and Computer Applications* (Elsevier). Submitted Mar. 2026; revised version Sep. 2026.
Under review — revision JCR Q1 IF 7.7

National Conferences

- [C1] Lara-Gutiérrez, A., Fernández-Gago, C. & Onieva, J. A. “HoneyV: laboratorio automatizado de análisis de malware basado en honeypots.” [HoneyV: an automated honeypot-based malware analysis laboratory] *XIX Reunión Española sobre Criptología y Seguridad de la Información (RECSI 2026)*, Universidad de La Laguna, Tenerife, 17–21 Mar. 2026. Proceedings, pp. 326–331.
- [C2] Lara-Gutiérrez, A., Fernández-Gago, C. & Onieva, J. A. “HDDAF: un framework para la detección y adaptación de drift en sistemas de detección de amenazas basados en inteligencia artificial.” [HDDAF: a framework for drift detection and adaptation in AI-based threat detection systems] *XI Jornadas Nacionales de Investigación en Ciberseguridad (JNIC 2026)*, Universitat Politècnica de Catalunya, Barcelona, 6–8 May 2026. Proceedings, pp. 88–89.

FELLOWSHIPS, GRANTS & AWARDS

Aug. 2026	Awarded Project — Chair of Cybersecurity, University of Málaga & Google (VirusTotal) <i>Competitive call of the UMA–Google Cybersecurity Chair</i> Project AgentAPT : an LLM agent with tool calling that interrogates Google Threat Intelligence / VirusTotal sandboxes to attribute malware samples to APT groups. Presented with the other awarded projects at the Google Safety Engineering Center (GSEC) Málaga, Sep. 2026.	Málaga, Spain
Current	FPU Predoctoral Fellowship (Formación de Profesorado Universitario) <i>Spanish Ministry of Science, Innovation and Universities</i> Highly competitive national doctoral contract. Host: Dept. of Languages and Computer Science, ETSI Informática, University of Málaga.	Spain
2022 – 2024	Extraordinary Graduation Award (Valedictorian) <i>School of Computer Engineering, University of Málaga</i> Highest academic record (ranked 1st) of the M.Sc. in Computer Engineering cohort.	Málaga, Spain
2018 – 2022	Extraordinary Graduation Award (Valedictorian) & Outstanding Thesis Award <i>School of Computer Engineering, University of Málaga</i> Highest academic record (ranked 1st) of the B.Sc. in Computer Engineering cohort.	Málaga, Spain
	Best Final Degree Project Award (2nd edition) <i>Chair of Commerce and Digital Transformation, University of Málaga</i> Recognises the best Bachelor’s/Master’s projects in digital transformation and consumer behaviour.	Málaga, Spain

RESEARCH EXPERIENCE

Feb. 2026 – Present	Visiting Researcher <i>University of Waikato — Te Ipu o Te Mahara (Artificial Intelligence Institute)</i> International research stay on AI security and intelligent systems at one of New Zealand’s leading AI research centres.	Hamilton, New Zealand
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Apr. 2024 – Present	R&D Engineer / Researcher <i>University of Málaga — NICS Lab (Network, Information and Computer Security)</i>	Málaga, Spain
	▶ Lead research on intelligent threat detection, proactive defense, honeypots, and AI-based malware classification and attribution. ▶ Contributing to a national public collaboration project with INCIBE (Spanish National Cybersecurity Institute).	
Mar. 2022 – Oct. 2023	Research & Technical Assistant <i>University of Málaga — Dept. of Languages and Computer Science</i>	Málaga, Spain
	▶ Built a private web area and multilingual editors (Django, MySQL, Linux) for the INMOCOR macrocorpus research project. ▶ Engineered a Java/Spring web application and MLP-regression visualisation tools.	

TEACHING & DISSEMINATION

2027	Invited Seminars — AIR Academy, AIR-Andalusia EDIH <i>European Digital Innovation Hub on AI & Robotics (University of Málaga)</i>	Málaga, Spain
	▶ <i>AI integration for anomaly detection</i> (2 h, co-delivered) — scheduled Apr. 2027. ▶ <i>AI integration for malware analysis</i> (2 h) — scheduled Sep. 2027.	

INDUSTRY EXPERIENCE

Jan. 2023 – Jan. 2024	Junior Integration Engineer <i>SoftProject Ibérica S.L.</i>	Málaga, Spain
	▶ Maintained and deployed applications on Windows and Linux for international enterprise clients. ▶ Developed demo web applications in CI/CD DevOps environments (XML/XSLT, JavaScript, SQL, JEE).	

SKILLS & LANGUAGES

Programming	Python, Java, C/C++ , JavaScript, R, Shell, CUDA, SQL, PHP
ML & AI	PyTorch, TensorFlow, scikit-learn, LightGBM, XGBoost, SHAP, Grad-CAM; GANs; LLMs (Gemma, LLaMA, Zephyr) and tool-calling agents
Security	Metasploit, Wireshark, Snort/Suricata, honeypots (OpenCanary), malware sandboxes, VirusTotal / GTI API
Research	Academic writing. Experimental design. Statistical testing. Peer review
Languages	Spanish (native). English (C2 — Cambridge Proficiency, score 207). French (B1)